NETFLOW

YERLİ ÜRETİM

SOLID MATERIAL · التحكم في المستوى · NT-POD2 SERIES

# مفتاح المستوى بالمحرك الدوار

Rotary paddle level sensor for full / empty detection in silos, bunkers and containers

The NT-POD2 series rotary paddle level switch mechanically detects the level of powder and granular solids in silos. The paddle, driven at 5 rpm by a geared synchronous motor, is stopped when covered by material; the resulting reaction movement of the motor body is detected by a limit switch (micro switch), a relay contact output is generated and the motor stops. Detection sensitivity can be set in 3 steps according to the bulk density of the material.

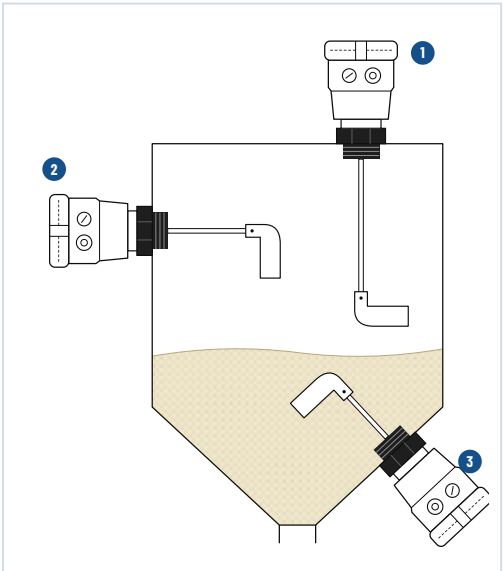


NT-POD2 · L TYPE PADDLE

| | |
|---|---|
| **IP66 / IP67**<br>درجة الحماية | **3 Steps**<br>SENSITIVITY SETTING |
| **24 VDC / 220 VAC**<br>SUPPLY OPTIONS | **5 rpm**<br>PADDLE SPEED |

- 100 / 200 mm shaft standard, 200 / 500 / 1000 mm extension shaft optional
- Limit switch / micro switch detection
- SS316 stainless steel shaft and paddle
- 1½" BSPP connection, 1" and 2" optional
- L and T type paddle, selected per application
- Glass-fibre reinforced PC-ABS connection; aluminium and stainless optional

## TYPICAL USE




**1 VERTICAL MOUNTING · HIGH LEVEL**

Mounted from the silo roof. Used for full-level alarm and filling stop; the switching point is set by the shaft length.

**2 SIDE MOUNTING · INTERMEDIATE LEVEL**

Mounted horizontally on the silo wall. For high or intermediate level control, positioned outside the filling stream.

**3 ANGLED MOUNTING · LOW LEVEL**

Mounted at an angle on the cone. Used for empty alarm and feed demand; the angled position reduces material build-up on the paddle.

## مجالات التطبيق



CEMENT · READY-MIX CONCRETE



GYPSUM · LIME · AGGREGATE

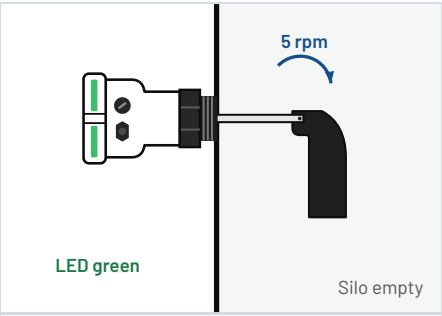


FEED · SEED · FLOUR



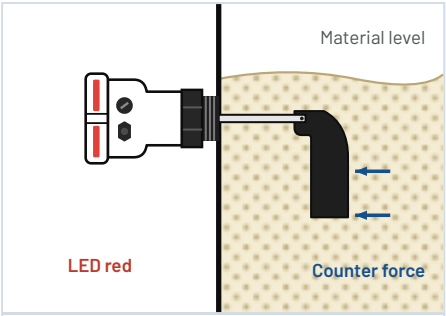
PLASTICS · COAL DUST

## OPERATING PRINCIPLE



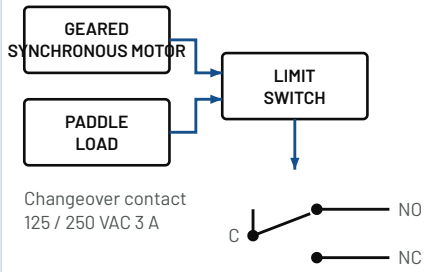

### 1 EMPTY STATE

The geared synchronous motor rotates the paddle shaft continuously at 5 rpm. As long as the shaft turns freely, the relay stays in the empty state and the indicator LED is green.




### 2 MATERIAL CONTACT

When material covers the paddle, a counter force acts on it. This force rises until it exceeds the threshold set by the 3-step spring adjustment.




### 3 SIGNAL OUTPUT

The load is detected by the limit switch; the motor stops and the changeover contact switches within max. 0.6 s, the indicator LED turns red. When the material recedes, the system resets itself.

## المواصفات الفنية

### ELECTRICAL DATA

| | |
|---|---|
| Supply (24 VDC) | 24 VDC ±3%, max. 2.7 W |
| Supply (220 VAC) | 220 VAC, max. 3.1 W |
| Signal output | 1 changeover contact (C / NO / NC), 24 VDC / 220 VAC 3 A |
| Signal delay | Max. 0.6 s |
| Terminal | Max. 2 mm² (AWG 14) cable |
| Electrical connection | PG 9 cable gland · optional DIN 43650 (ABS) connector or M12 4-pin |
| Status indicator | LED · empty: green · full: red |
| درجة الحماية | IP66 / IP67 |

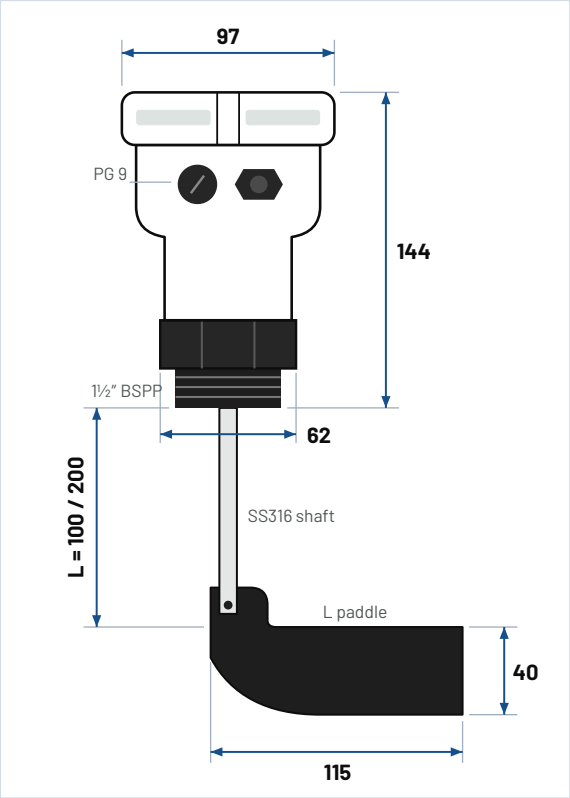
### OPERATING CONDITIONS

| | |
|---|---|
| Ambient temperature | –20 °C ... +80 °C |
| Process temperature | –5 °C ... +80 °C · +160 °C with cooling extension |
| Silo pressure | –0,5 ... +0,5 bar |
| Max. operating vibration | 5 – 500 Hz, 3G RMS (IEC 60068-2-64) |

### MECHANICAL DATA

| | |
|---|---|
| Housing | Option P: PC-ABS · Option A: machined aluminium |
| Drive | Geared synchronous motor |
| Detection | Limit switch (micro switch) |
| Shaft bearing | Double ball bearing |
| Sealing | NBR seal (Viton optional) |
| Connection material | Glass-fibre reinforced PC-ABS · aluminium and stainless optional |
| Connection size | 1½" BSPP (standard) · 1" and 2" BSPP optional |
| Shaft and paddle | Stainless steel SS316 |
| Shaft length | 100 mm (standard) · 200 mm · optional 200 / 500 / 1000 mm extension shaft |
| Shaft speed | 5 rpm |
| Sensitivity setting | 3-step spring |
| Min. bulk density | 250 g/l (with standard L paddle) |
| Weight | 870 g (100 mm shaft) · 1000 g (200 mm shaft), excl. extension shafts |

## الأبعاد الأبعاد mm

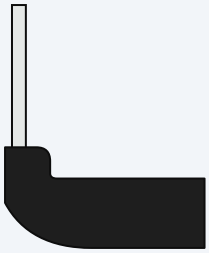



## CONNECTION SIZES

| 1" BSPP | 1½" BSPP | 2" BSPP |
|---|---|---|
| OPTIONAL | STANDARD | OPTIONAL |

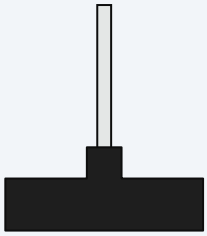
Shaft length 100 mm (standard) or 200 mm. For deeper switching points, 500 / 1000 mm extension shafts are available as options.

## PADDLE TYPES



### L TYPE

Standard. Powders and fine granules, density above 250 g/l.



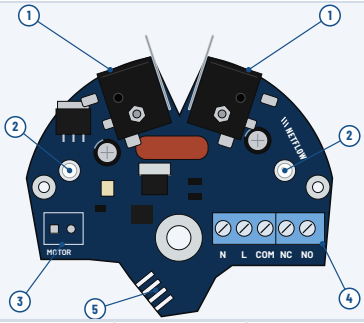
### T TYPE

Large surface. Earlier detection in light and fluffy materials. Single blade, rod-connected spring system.

## ELECTRICAL CONNECTION ELECTRONIC BOARD

### 220 VAC MODEL

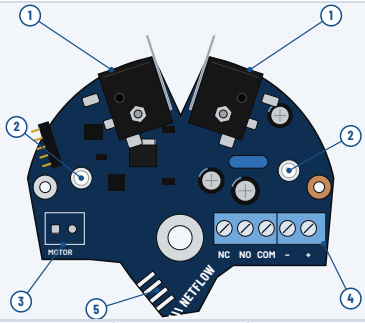
Supply: terminals 1 – 2




| N | L | COM | NC | NO |
|---|---|---|---|---|
| 1 · Neutral | 2 · Phase | 3 · Common | 4 · N. closed | 5 · N. open |

### 24 VDC MODEL

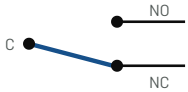
Supply: terminals 4 – 5



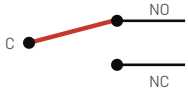

| NC | NO | COM | – | + |
|---|---|---|---|---|
| 1 · N. closed | 2 · N. open | 3 · Common | 4 · 0 V | 5 · 24 VDC |

① **Limit switch** 2 pcs, load detection ② **LED** Empty: green · Full: red ③ **Motor** Motor connection ④ **Terminal block** 5-pole, dry contact ⑤ **Spring step** Torque setting

### SIGNAL STATUS




**SILO EMPTY**
C – NC closed · LED green · paddle rotates



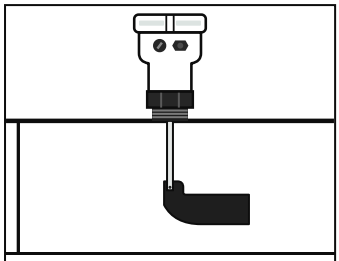

**SILO FULL**
C – NO closed · LED red · motor stops

### CONNECTION TYPES

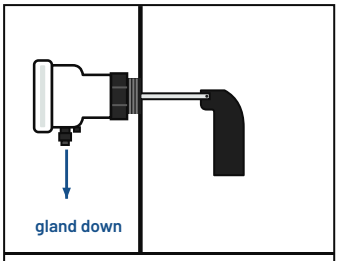
| | |
|---|---|
| **PG 9 cable gland** | Standard |
| **DIN 43650 (ABS) connector** | Optional |
| **M12 4-pin connector** | Optional |

Electrical connections must be made by a qualified electrician with the power off. Max. 2 mm² cable can be connected to the terminals.

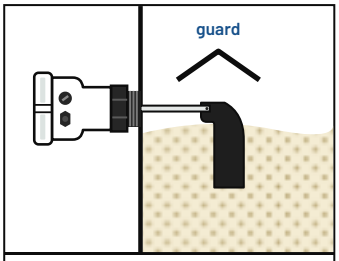
## MECHANICAL التركيب



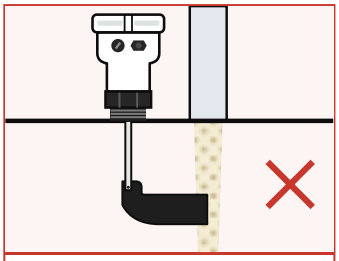
✓ Vertical from roof, perpendicular to surface




✓ In horizontal mounting the gland faces down




✓ At low level the paddle is protected by a guard



✗ Do not mount below the filling stream

**01** The unit must be mounted perpendicular to the surface, away from the material inlet and filling stream.

**02** The paddle must not touch the silo wall, and material must not jam between them.

**03** In horizontal mounting, the cable gland or connector must face down to prevent water ingress.

**04** In low-level applications and with heavy materials, a guard must be fitted above the paddle.

**05** After installation, tighten the housing cover and gland fully to ensure sealing.

**06** Do not mount on vibrating cones or vibrating surfaces.

## TROUBLESHOOTING

| SYMPTOM | POSSIBLE CAUSE | REMEDY |
|---|---|---|
| **Shaft not rotating, LED off** | No supply or wrong wiring | Measure the supply voltage at terminals 1 – 4 for 24 VDC and 1 – 4 for 220 VAC with PG 9 connection; at the relevant pins for M12 and DIN connectors. |
| **Full signal with empty silo** | Material jammed between paddle and silo wall | Clean the paddle, increase the mounting distance or add a guard. |
| **No signal with material present** | Material density below 250 g/l or spring step too high | Lower the spring step according to material density; if needed, change to a T type paddle. |
| **Moisture / water in housing** | Gland facing up or cover not fully closed | Turn the gland downwards, check the cover and seal. |
| **Signal flickering** | Mounted on a vibrating surface | Move the unit away from the vibrating cone. |

## SELECTION GUIDE

### HOUSING

A plastic housing is sufficient for indoor and normal dusty applications. For outdoor, wash-down or impact-risk lines, an aluminium housing is preferred.

### SHAFT LENGTH AND PADDLE

For materials that stick to the silo wall, choose a 200 mm shaft or an extension shaft. For light materials below 250 g/l, use a T type paddle.

### HOT PROCESSES

Where the process temperature exceeds +80 °C, use the model with cooling extension (up to +160 °C). Stainless steel is recommended as connection material.

## OPTIONS

- Aluminium housing (machined)
- Cooling extension model, +160 °C process
- 200 / 500 / 1000 mm extension shaft
- T type paddle
- 1" and 2" BSP connection
- Aluminium or stainless connection
- DIN 43650 (ABS) or M12 4-pin connector
- Viton seal

### COMPLIANCE

ISO 9001 · EN 61000-6-4 · EN 61000-6-2 · EN 61010-1

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Contact our technical team to select the right shaft length, paddle type and connection size for your application.

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